

Development of a Compartmentalized Metabolic Subnetwork for the Endothelial Cytoskeleton to Model SHock-Induced Endotheliopathy (SHINE)

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1. **Introduction: The Biological and Modeling Problem**

1.1. **Pathophysiology of SHock-Induced Endotheliopathy (SHINE)**

SHock-Induced Endotheliopathy (SHINE) is a critical condition where systemic shock leads to widespread injury of the endothelial cells lining blood vessels. This "endotheliopathy" is characterized by the breakdown of the endothelial barrier, leading to vascular leakage, edema, and subsequent multiple organ failure. The molecular drivers of SHINE include a massive inflammatory response and sympatho-adrenal hyperactivation.

1.2. **The Cytoskeleton as a Central Player in Endothelial Dysfunction**

The integrity of the endothelial barrier is physically maintained by the cell's cytoskeleton—a dynamic network of protein filaments. In SHINE, endothelial cells undergo a dramatic pathological reorganization of this network, shifting from a stable state to a hyper-contractile state characterized by the formation of actin stress fibers. This process actively pulls cells apart, creating intercellular gaps and driving vascular leakage.

1.3. **The Modeling Objective: Transcending the Limitations of Static Representations**

The main goal was to develop a metabolic model capable of capturing the unique bioenergetic demands of the endothelial cytoskeleton during SHINE. Initial approaches in genome-scale models (GEMs) often use "demand" reactions to represent the cost of building cellular components. However, the pathology of SHINE is not simply the synthesis of *more* cytoskeleton, but the frantic, energy-intensive *remodeling* of the existing structure. This required a move from a static model of composition to a dynamic model of function.

2. **Phase 1: The Initial Approach – A Single Aggregated Demand Reaction**

2.1. **Theoretical Basis: The Role of Demand Reactions in GEMs**

The first logical step was to create a single demand reaction. In a GEM, this is a sink reaction that consumes a stoichiometrically defined set of precursors (amino acids, ATP, etc.) to represent the production of a specific macromolecule or the entire cell biomass.

2.2. **Quantitative Composition of the Structural Cytoskeleton**

To build this reaction, we first defined the bulk composition of the primary structural filaments of the cytoskeleton.

Table 2.1: Initial 3-Component Mass Distribution

Component	Mass Percentage	Primary Protein(s)	UniProt ID(s) Used
:---	:---	:---	:---
Actin Filaments	50%	Actin, cytoplasmic 1	P60709

| **Microtubules** | 30% | Tubulin alpha-1A & Tubulin beta | P68363, P07437 |
 | **Intermediate Filaments** | 20% | Vimentin | P08670 |

2.3. Derivation and Formulation of the Initial Demand Reaction

Based on the weighted average of the amino acid compositions from the proteins listed in Table 2.1, and including the energy cost for polymerization (ATP for actin, GTP for tubulin), the following single demand reaction was formulated to represent the synthesis of 1 gram of cytoskeleton.

Table 2.2: Detailed Stoichiometry of the Aggregated Reaction

| Metabolite | mmol required per 1g Cytoskeleton |

| :--- | :--- |

| L-Alanine | 0.776 |

| L-Arginine | 0.435 |

| L-Asparagine | 0.389 |

| L-Aspartate | 0.814 |

| L-Cysteine | 0.222 |

| L-Glutamine | 0.403 |

| L-Glutamate | 1.056 |

| Glycine | 0.867 |

| L-Histidine | 0.224 |

| L-Isoleucine | 0.540 |

| L-Leucine | 0.825 |

| L-Lysine | 0.505 |

| L-Methionine | 0.218 |

| L-Phenylalanine | 0.380 |

| L-Proline | 0.463 |

| L-Serine | 0.627 |

| L-Threonine | 0.518 |

| L-Tryptophan | 0.106 |

| L-Tyrosine | 0.301 |

| L-Valine | 0.655 |

| ATP (+ H₂O) | 5.86 |

| GTP (+ H₂O) | 2.94 |

3. Phase 2: The Critical Turning Point – Identifying the Model's Limitations

3.1. The Fundamental Flaw: Conflating Static Synthesis with Dynamic Remodeling

A critical analysis revealed that the single demand reaction was fundamentally unsuitable for modeling SHINE. It rigidly links the energy cost (ATP/GTP) to the synthesis of new protein mass (amino acids). However, the pathophysiology of SHINE is dominated by a massive increase in energy consumption for the *reorganization* of the *existing* protein pool (e.g., actin treadmilling, actomyosin contraction). This remodeling process consumes energy at a rate independent of, and far greater than, new protein synthesis.

3.2. The Pathophysiological Imperative: Decoupling Energy Costs for SHINE

The key insight was the need to decouple these two distinct metabolic burdens. A valid model must be able to independently simulate:

1. **The Cost of Synthesis:** The drain of amino acids and a baseline ATP cost for creating new proteins to replace damaged ones.
2. **The Cost of Remodeling:** A separate, variable, and much larger energy drain (ATP/GTP hydrolysis) representing the hyper-dynamic state of the cytoskeleton during SHINE.

3.3. Conclusion: Abandoning the Single Reaction for a Subnetwork Approach

The single demand reaction was discarded in favor of a more sophisticated framework that could treat the cytoskeleton as a dynamic, energy-consuming subsystem.

4. Phase 3: Design of the Advanced Subnetwork

4.1. Conceptual Framework: A Virtual Cytoskeleton Compartment ([ck])

The chosen solution was to create a new, virtual "cytoskeleton" compartment ([ck]) within the model. This design provides modularity, allows for the representation of local metabolite pools, and is extensible for future additions.

4.2. Refining the Framework: The "Local Synthesis" Model

The most advanced model design, and the one we adopted, was the "**local synthesis**" model. Instead of synthesizing proteins in the cytosol ([c]) and transporting the finished products, this model transports the basic building blocks (amino acids, ATP) into the [ck] compartment and simulates protein synthesis *in situ*. This reflects biological evidence for localized translation of cytoskeletal components and offers the most granular control.

4.3. Expanding Biochemical Scope: Incorporating Regulatory Glycans and Lipids

To increase the model's biological realism, key regulatory molecules with direct metabolic links were incorporated:

- **Glycans (O-GlcNAcylation):** The substrate **UDP-GlcNAc**, derived from glucose, was added to model the post-translational modification of proteins like vimentin.
- **Lipids (PIP2):** **Phosphatidylinositol 4,5-bisphosphate (PIP2)** was included as a key regulator of actin dynamics at the plasma membrane.

5. Phase 4: Curation, Refinement, and Data Integration

5.1. Analysis of External Metabolic Data

A list of potential metabolites and reactions for the [ck] compartment was analyzed to identify which processes were truly localized to the cytoskeleton versus being part of general cytosolic metabolism.

Table 5.1: Analysis of Metabolites from compartment_ck_metabolites.csv

Metabolite	ID	Include in [ck]?	Justification
:---	:---	:---	:---

ATP, ADP, GTP, GDP, Pi, H+, H2O Various Yes Direct substrates/products for polymerization and contraction.
All Amino Acids Various Yes Substrates for local synthesis.
[myosin light chain] MAM00190ck Yes Direct substrate for the contractile machinery.
PIP2 Precursors MAM00553ck Yes Local synthesis of the lipid regulator PIP2.
Nitric oxide MAM02609ck No Diffusible signaling molecule produced in the cytosol.
Glucose, Succinate, Zymosterol Various No Part of central metabolism in other compartments (cytosol, mitochondria).

Table 5.2: Analysis of Reactions from compartment_ck_reactions.csv

Reaction	ID	Include in [ck]?	Justification
:--- :--- :--- :---			
myosin light chain + ATP --> ...	MAR22907	Yes	The core reaction of actomyosin contraction.
ADP + GTP <=> ATP + GDP	MAR13466	Yes	Local regeneration of GTP for tubulin polymerization.
ATP + PI(4)P --> PIP2 + ADP	MAR16740	Yes	Local synthesis of the regulatory lipid PIP2.
L-Arginine + ... --> Nitric oxide	MAR23221	No	Nitric Oxide Synthase (NOS) is a cytosolic enzyme.
Branched-chain aa catabolism	MAR18983	No	This is a mitochondrial pathway.
Fatty acid synthesis	MAR14567	No	This is a cytosolic pathway.

5.2. Iterative Debugging: Identifying and Resolving Blocked Reactions

The initial subnetwork design was found to contain blocked reactions because several byproducts were "dead-end" metabolites with no exit pathway.

Table 5.3: Identified Dead-End Metabolites and Blocked Reactions

Dead-End Metabolite	Produced by Reactions...
:--- :---	
pi[ck] (Phosphate)	Actin_Remodeling, Tubulin_Remodeling, Myosin_Dephosphorylation
h[ck] (Proton)	Actin_Remodeling, Tubulin_Remodeling
udp[ck] (Uridine diphosphate)	Vimentin_Glycosylation, Actin_Modification

To resolve this, reversible export transport reactions were added.

Table 5.4: Corrective Export Reactions to Unblock the Network

Reaction ID	Reaction Equation
:--- :---	
T_Pi_out	pi[ck] <=> pi[c]
T_H_out	h[ck] <=> h[c]
T_UDP_out	udp[ck] <=> udp[c]

5.3. Final Integration: The Myosin Contractile Machinery

A final curation step identified that the myosin light chain was being treated as an external input, inconsistent with the local synthesis model. To correct this, the transport of the pre-made protein was removed, and a new local synthesis reaction was added.

Table 5.5: Data for Myosin Light Chain 9 Synthesis

Protein Myosin light polypeptide 9 (MYL9)

| :--- | :--- |
 | **UniProt ID** | P24844 |
 | **Length** | 172 amino acids |
 | **Translation Cost** | 684 ATP |

5.4. Refinement of the Contractile Mechanism: The Coupled "Event" Model

5.4.1. Modeling Contraction as a Unified Event

To accurately simulate the energy dissipation of SHINE, the model must ensure that every signaling event triggers the associated metabolic cost. A potential risk in subnetwork design is "short-circuiting," where the solver finds a low-cost path (e.g., simple phosphorylation) that satisfies stoichiometric constraints without performing the high-cost mechanical work.

To eliminate this possibility, we modeled the contractile process as a **unified, coupled reaction**. Instead of separating activation (phosphorylation) from mechanical cycling, we combined them into a single "Contractile Event."

5.4.2. The "Contractile Event" Topology

In this finalized design, the reaction represents the aggregate sum of activating the myosin head and running it through approximately 150 mechanical cycles.

1. **The Input:** Resting Myosin, Cytosolic Calcium (Signal), and 151 ATP (1 for activation + 150 for cycling).
2. **The Output:** Phosphorylated Myosin, Cytoskeletal Calcium (Transmission), and the hydrolysis byproducts.
3. **The Reset:** A separate reaction dephosphorylates the myosin, returning it to the resting pool to be used again.

This topology forces a strict dependency: The calcium signal cannot be processed without paying the full metabolic cost of 151 ATP.

Table 5.6: Final Specification for Contractile Reactions

Reaction ID	Reaction Equation	Role
R_Myosin_Event	$\text{myosin_light_chain}[\text{ck}] + 151 \text{ atp}[\text{ck}] + 150 \text{ h}_2\text{o}[\text{ck}] + \text{ca}_2[\text{c}] \rightarrow \text{myosin_light_chain_P}[\text{ck}] + 151 \text{ adp}[\text{ck}] + 150 \text{ pi}[\text{ck}] + 151 \text{ h}[\text{ck}] + \text{ca}_2[\text{ck}]$	The Unified Event: Combines activation and work. Forces the consumption of 151 ATP per calcium signaling unit, preventing metabolic bypass.
T_Ca	$\text{ca}_2[\text{ck}] \rightleftharpoons \text{ca}_2[\text{c}]$	Signal Loop: Allows the calcium to cycle back to the cytosol, completing the flux loop.
R_Myosin_Reset	$\text{myosin_light_chain_P}[\text{ck}] + \text{h}_2\text{o}[\text{ck}] \rightarrow \text{myosin_light_chain}[\text{ck}] + \text{pi}[\text{ck}]$	System Reset: Dephosphorylates the myosin, making it available for the next contractile event.

6. Phase 5: Finalizing the Model Composition

6.1. Revisiting the Mass Distribution: Incorporating Myosin

With the integration of myosin, the initial 3-component composition was re-evaluated to create a more comprehensive 4-component model that includes the contractile machinery. Myosin II was estimated to contribute ~4% of the total mass of the core cytoskeletal system.

6.2. The Final Reference Composition for Flux Analysis

This revised composition serves as the definitive baseline for setting realistic fluxes in the final, flexible subnetwork model.

Table 6.1: Revised 4-Component Cytoskeletal Mass Distribution

Component	Revised Percentage	Role
Actin Filaments	48.0%	Structural Polymer (Scaffold)
Microtubules (Tubulin)	28.8%	Structural Polymer (Tracks)
Intermediate Filaments (Vimentin)	19.2%	Structural Polymer (Mechanical Strength)
Myosin II Motor Protein	4.0%	Motor Protein (Force Generation)
Total	100.0%	

7. Final Model Specification: The Complete Reaction Set

This is the complete, final list of all reactions in the cytoskeleton subnetwork.

7.1. Block 1: Substrate Transport (from [c] to [ck])

26 reversible reactions

Reaction ID	Reaction Equation
T_Ala to T_Val	(amino_acid)[c] <=> (amino_acid)[ck] (20 reactions)
T_ATP	atp[c] <=> atp[ck]
T_GTP	gtp[c] <=> gtp[ck]
T_UDP_GlcNAc	udp_glcna[c] <=> udp_glcna[ck]
T_PI4P	pi4p[c] <=> pi4p[ck]
T_H2O	h2o[c] <=> h2o[ck]
T_Ca	Ca2[c] <=> Ca2[ck]

7.2. Block 2: Macromolecule Synthesis (in Cytoskeleton [ck])

4 irreversible reactions

Reaction ID	Reaction Equation
-------------	-------------------

| :--- | :--- |

| **R_Synth_Actin** | 22 ala-L[ck] + 18 arg-L[ck] + 15 asn-L[ck] + 30 asp-L[ck] + 16 cys-L[ck] + 24 gln-L[ck] + 26 glu-L[ck] + 28 gly[ck] + 8 his-L[ck] + 29 ile-L[ck] + 26 leu-L[ck] + 20 lys-L[ck] + 16 met-L[ck] + 15 phe-L[ck] + 17 pro-L[ck] + 21 ser-L[ck] + 27 thr-L[ck] + 4 trp-L[ck] + 15 tyr-L[ck] + 18 val-L[ck] + 1496 atp[ck] + 1122 h2o[ck] --> actin_protein_bare[ck] + 1496 adp[ck] + 1496 pi[ck] + 1496 h[ck] |

| **R_Synth_Tubulin** | 61 ala-L[ck] + 40 arg-L[ck] + 37 asn-L[ck] + 62 asp-L[ck] + 23 cys-L[ck] + 35 gln-L[ck] + 83 glu-L[ck] + 74 gly[ck] + 22 his-L[ck] + 46 ile-L[ck] + 71 leu-L[ck] + 42 lys-L[ck] + 22 met-L[ck] + 42 phe-L[ck] + 46 pro-L[ck] + 62 ser-L[ck] + 55 thr-L[ck] + 12 trp-L[ck] + 36 tyr-L[ck] + 65 val-L[ck] + 3580 atp[ck] + 2685 h2o[ck] --> tubulin_dimer_bare[ck] + 3580 adp[ck] + 3580 pi[ck] + 3580 h[ck] |

| **R_Synth_Vimentin** | 36 ala-L[ck] + 25 arg-L[ck] + 21 asn-L[ck] + 37 asp-L[ck] + 6 cys-L[ck] + 28 gln-L[ck] + 53 glu-L[ck] + 20 gly[ck] + 7 his-L[ck] + 20 ile-L[ck] + 44 leu-L[ck] + 23 lys-L[ck] + 8 met-L[ck] + 13 phe-L[ck] + 17 pro-L[ck] + 37 ser-L[ck] + 26 thr-L[ck] + 3 trp-L[ck] + 16 tyr-L[ck] + 26 val-L[ck] + 1860 atp[ck] + 1395 h2o[ck] --> vimentin_protein_bare[ck] + 1860 adp[ck] + 1860 pi[ck] + 1860 h[ck] |

| **R_Synth_MyosinLC** | 15 ala-L[ck] + 5 arg-L[ck] + 9 asn-L[ck] + 14 asp-L[ck] + 1 cys-L[ck] + 6 gln-L[ck] + 19 glu-L[ck] + 10 gly[ck] + 2 his-L[ck] + 8 ile-L[ck] + 10 leu-L[ck] + 14 lys-L[ck] + 6 met-L[ck] + 8 phe-L[ck] + 6 pro-L[ck] + 7 ser-L[ck] + 8 thr-L[ck] + 0 trp-L[ck] + 3 tyr-L[ck] + 8 val-L[ck] + 684 atp[ck] + 513 h2o[ck] --> myosin_light_chain[ck] + 684 adp[ck] + 684 pi[ck] + 684 h[ck] |

7.3. Block 3: Local Regulation & Protein Modification (in Cytoskeleton [ck])

4 reactions

| Reaction ID | Reaction Equation |

| :--- | :--- |

| **R_PIP2_Synth** | pi4p[ck] + atp[ck] --> pip2[ck] + adp[ck] + h[ck] |

| **R_GTP_Regen_NDPK** | gdp[ck] + atp[ck] <=> gtp[ck] + adp[ck] |

| **R_Vim_Glyco** | vimentin_protein_bare[ck] + udp_glcna[ck] --> vimentin_GlcNAc[ck] + udp[ck] |

| **R_Actin_Mod** | actin_protein_bare[ck] + udp_glcna[ck] + pip2[ck] --> actin_complex[ck] + udp[ck] |

7.4. Block 4: Energy Dissipation & Cytoskeletal Function (in Cytoskeleton [ck])

3 irreversible reactions

| Reaction ID | Reaction Equation |

| :--- | :--- |

| **R_Myosin_Contract** | myosin_light_chain[ck] + 151 atp[ck] + 150 h2o[ck] + ca2[c] --> myosin_light_chain_P[ck] + 151 adp[ck] + 150 pi[ck] + 151 h[ck] + ca2[ck] |

| **R_Actin_Remodel** | atp[ck] + h2o[ck] --> adp[ck] + pi[ck] + h[ck] |

| **R_Tubulin_Remodel** | gtp[ck] + h2o[ck] --> gdp[ck] + pi[ck] + h[ck] |

7.5. Block 5: Product Turnover & Waste Export (in and out of [ck])

10 reactions

| Reaction ID | Reaction Equation |

```

| :--- | :--- |
| R_Myosin_Dephospho | myosin_light_chain_P[ck] + h2o[ck] --> myosin_light_chain[ck] +
pi[ck] |
| DM_Actin | actin_complex[ck] --> |
| DM_Tubulin | tubulin_dimer_bare[ck] --> |
| DM_Vimentin | vimentin_GlcNAc[ck] --> |
| DM_MyosinLC | myosin_light_chain[ck] --> |
| T_ADG_out | adp[ck] <=> adp[c] |
| T_GDP_out | gdp[ck] <=> gdp[c] |
| T_Pi_out | pi[ck] <=> pi[c] |
| T_H_out | h[ck] <=> h[c] |
| T_UDP_out | udp[ck] <=> udp[c] |

```

8. Conclusion: A Mechanistically Detailed and Robust Modeling Tool

The process documented in this report has yielded a compartmentalized metabolic subnetwork of the endothelial cytoskeleton. By progressing from a simple demand reaction to this final, integrated model, we have created a tool that is mechanistically suited for studying the complex pathophysiology of SHINE. The final model correctly decouples the metabolic costs of protein synthesis from the massive energy demands of dynamic remodeling and contraction, incorporates key regulatory inputs from glucose and lipid metabolism, and is structurally unblocked. This subnetwork now serves as a robust platform for generating testable hypotheses about the metabolic vulnerabilities of endothelial cells cytoskeleton during critical illness.

9. References and Data Sources

1. **UniProt Consortium.** (2023). UniProt: the universal protein knowledgebase in 2023. *Nucleic Acids Research*, 51(D1), D523–D531.
 - **Relevance:** Authoritative source for the primary amino acid sequences used to determine the exact stoichiometric requirements for the synthesis reactions of Actin (P60709), Tubulin alpha-1A (P68363), Tubulin beta (P07437), Vimentin (P08670), and Myosin Light Chain 9 (P24844).
2. **Thiele, I., & Palsson, B. Ø.** (2010). A protocol for generating a high-quality genome-scale metabolic reconstruction. *Nature Protocols*, 5(1), 93–121.
 - **Relevance:** Provided the standard methodology for constructing macromolecule synthesis reactions from primary sequence data and for accounting for energy costs, which guided the entire model-building process.
3. **Johansson, P. I., Stensballe, J., & Ostrowski, S. R.** (2017). Shock induced endotheliopathy (SHINE) in acute critical illness - a unifying pathogenic mechanism. *Critical Care*, 21(1), 25.
 - **Relevance:** Formally defined SHINE, providing the essential biological and clinical context for our modeling objective.
4. **Ince, C., et al.** (2016). The endothelium in sepsis. *Shock*, 45(3), 259–270.

- **Relevance:** Justified the focus on *remodeling* of the actin cytoskeleton (stress fiber formation) as the key pathological and energy-consuming event in endothelial dysfunction.
5. **Dudek, S. M., & Garcia, J. G. N.** (2001). Cytoskeletal regulation of pulmonary vascular permeability. *Journal of Applied Physiology*, 91(4), 1487–1500.
 - **Relevance:** Provided the specific biochemical justification for creating high-flux energy dissipation reactions, detailing the ATP-dependent nature of actomyosin contraction.
 6. **Hart, G. W., et al.** (2011). Cross talk between O-GlcNAcylation and phosphorylation: roles in signaling, transcription, and disease. *Annual Review of Biochemistry*, 80, 825–858.
 - **Relevance:** Justified the inclusion of O-GlcNAcylation by identifying cytoskeletal proteins as key targets of this nutrient-sensing modification, linking glucose metabolism to cytoskeletal function.
 7. **Yin, H. L., & Janmey, P. A.** (2003). Phosphoinositide regulation of the actin cytoskeleton. *Annual review of physiology*, 65, 761-789.
 - **Relevance:** Provided the rationale for including lipid metabolism by detailing the critical role of PIP2 in regulating actin-associated proteins at the cell membrane.